

TelcoPulse AI: A Hybrid Predictive and Generative Solution for Telecom Subscriber Churn

AI Solution Design and Prototype Evaluation

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Live demo: <https://telcopulse-ai.vercel.app> Source code: <https://github.com/mn3265-commits/telcopulse-ai>

1. Problem Definition and Context

1.1 Problem statement

Indonesian mobile operators face structural churn pressure from SIM consolidation, aggressive prepaid pricing competition, and a rising base of dual-SIM consumers. Indosat Ooredoo Hutchison (IOH), the country's second-largest operator, serves approximately 95 million subscribers and reports a blended monthly churn rate in the 3–4% range. Even at the lower bound, this implies roughly 3.3 million subscribers leave each month, a material revenue exposure given a blended ARPU near \$8.50 per month (IOH, 2024).

The problem this project addresses is narrowly scoped: **identify, ahead of time, which prepaid and postpaid subscribers are most likely to churn within the next 30 days, and equip the customer-value-management (CVM) team with the content needed to act on that signal.** The scope is deliberately not enterprise-wide transformation; it focuses on one decision (who to contact this month for retention), one model (a churn classifier), and one execution surface (multi-channel campaign content).

1.2 Current state

The CVM workflow at most regional operators today is reactive. Customer service teams receive inbound complaints, deactivation requests, or port-out notifications, then attempt to retain subscribers after the decision to leave is already partly made. There is no shared subscriber-level risk score across teams; segmentation is performed in batch via SQL by analysts on request, with multi-day turnaround; and campaign content is hand-drafted per channel, leading to inconsistent tone and missed channels. The result is that the highest-leverage retention window — the days *before* a subscriber attempts to switch — is largely unused.

1.3 Definition of success

A successful AI-enabled solution must, at minimum, do four things:

1. **Predict** a per-subscriber churn probability within a 30-day horizon, at sub-second latency, with a model that materially outperforms a rule-based heuristic on the same data.
2. **Translate** plain-language segment intent (for example, "high-value postpaid users with declining usage") into an executable filter without requiring SQL knowledge.
3. **Generate** on-brand, channel-specific retention content (SMS, email, push, WhatsApp) from a single brief.
4. **Quantify** the expected reach, conversion, and revenue impact of any campaign before it is sent, so a CVM lead can decide whether to proceed.

These four capabilities collapse what is currently a multi-team, multi-day workflow into a single dashboard interaction.

2. Data and AI Factory Design

2.1 Data sources and structure

The prototype uses a **synthetic dataset of 10,000 subscribers** generated by `scripts/generate_dataset.py`. The schema is modeled on the actual subscriber data structure used in CVM at IOH and Smartfren — internal CRM (demographics, plan), CDR (voice/SMS/data usage), billing (topups, monthly spend), customer service (complaints, support tickets, NPS), and network telemetry (average speed, dropped calls). Each record has 25 fields, of which 18 are used as model features.

Although the data is synthetic, every signal pattern reflects production reality: prepaid/postpaid plan share is set at 73/27, matching the Indonesian market split; city distribution is weighted by actual Indonesian demographics; usage and spend correlate with age and plan tier; and the churn label is generated from a documented combination of behavioral risk and protective factors with realistic noise.

2.2 Volume, velocity, and quality

- **Volume.** The prototype operates at 10,000 subscribers — a deliberately small slice that exercises every component of the pipeline end-to-end. At production scale, the same pipeline must handle ~95M subscribers monthly, implying tens of GB of CDR rows per day. The architectural choices below (batch scoring, columnar feature joins, serialized model artifact) anticipate this.
- **Velocity.** Behavioral features change daily (usage, topups, app logins). Identity and plan features change rarely. The pipeline assumes a daily batch refresh of behavioral features and a weekly model retrain, both of which are well within current operator data-engineering capacity.
- **Quality.** Real telecom data has three known quality issues: (a) missing NPS for non-respondents (~70% missing in production), (b) duplicated subscriber IDs after SIM swaps and re-registration drives, and (c) short tenure misreporting after the 2018 SIM re-registration mandate. The prototype dataset is clean by

construction; the production rollout plan in §5 explicitly calls for a data-quality sprint to address these before scaled deployment.

2.3 Data preparation

Preparation is performed once per training run by `scripts/train_model.py`. Steps:

1. Load the raw subscriber CSV.
2. Engineer three derived features: `usage_ratio` (data used ÷ quota), `engagement_score` (app logins ÷ 30 days), `is_prepaid` (one-hot for plan type).
3. Stratified 80/20 train/test split (8,000 / 2,000) with a fixed random seed for reproducibility.
4. For the logistic regression baseline only: standardize features with `StandardScaler` before fitting.

No imputation is needed on the synthetic data, but the production plan adds median-imputation for NPS, last-known-value carry-forward for missing CDR days, and a `nps_missing` indicator feature.

2.4 AI factory components

The pipeline implements all six components required by the AI factory pattern:

#	Component	Implementation in this project
1	Ingestion & preprocessing	CSV load + three engineered features (<code>scripts/train_model.py</code>)
2	Feature engineering	18 features chosen for behavioral, demographic, and network coverage
3	Model training	XGBoost classifier with class balancing; LR and rule-based baselines for comparison
4	Evaluation & validation	Held-out test split, PR-curve threshold tuning, confusion matrix, baseline comparison
5	Deployment / inference	Two-tier inference. ML scoring is batch (XGBoost runs over the full subscriber CSV at training time and persists <code>subscribers_scored.csv</code>); generative steps are on-demand real-time through seven Next.js API routes (<code>/api/churn</code> , <code>/api/segment</code> , <code>/api/campaign</code> , <code>/api/insights</code> , <code>/api/persona</code> , <code>/api/launch</code> , <code>/api/subscriber-brief</code>), each invoking Claude with a strict JSON schema and a deterministic template fallback when no API key is set. Two further outbound endpoints (<code>/api/send-email</code> , <code>/api/send-call</code>) deliver retention messages through Gmail SMTP and Twilio Voice.
6	Feedback loop	The Subscriber Workflow module captures per-subscriber human decisions (Approve / Escalate / Mark Safe / Outcome / reviewer notes) and submits them through <code>/api/feedback</code> , which logs the row for the next weekly retrain. In production this sink would be a Postgres / Supabase table consumed by the training job; the prototype acknowledges + logs to serverless function output so the loop is observable end-to-end without a database

2.5 Pipeline diagram

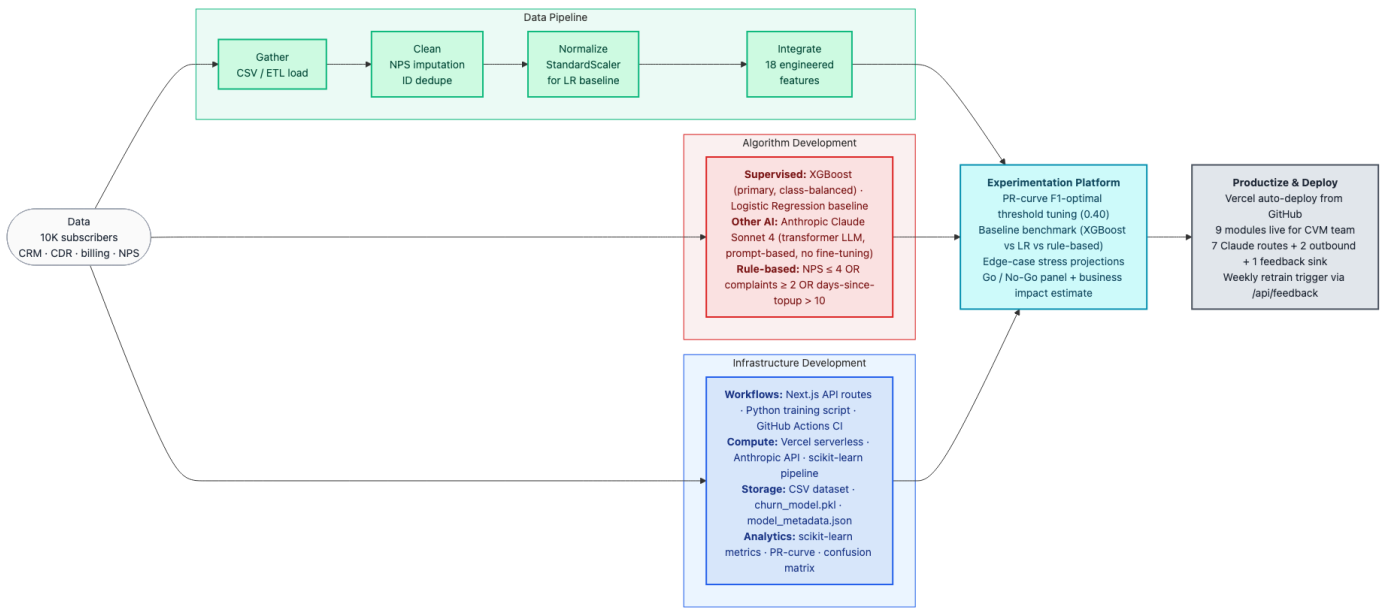


Figure 1. AI Factory pattern applied to TelcoPulse: three parallel tracks — **Data Pipeline** (Gather → Clean → Normalize → Integrate), **Algorithm Development** (supervised XGBoost + LR baselines · transformer-based Claude Sonnet 4 prompt-only · rule-based heuristic), and **Infrastructure Development** (workflows, compute, storage, analytics) — all converging into the **Experimentation Platform** (PR-curve tuning, baseline benchmark, edge cases, Go/No-Go) and then **Productize & Deploy** on Vercel with the weekly retrain trigger via `/api/feedback`.

3. AI Techniques and Technology Stack

3.1 AI techniques and justification

The solution is intentionally hybrid because no single AI paradigm fits all four required capabilities.

Predictive ML (gradient-boosted trees, supervised classification). Churn scoring is a binary classification problem on tabular, mixed-type features with strong univariate signals (NPS, complaints, days-since-topup) and meaningful interactions (tenure × plan type). On data of this shape, gradient-boosted trees consistently match or beat deep networks while being faster to train, easier to interpret, and trivial to serve. The prototype uses **XGBoost** with `scale_pos_weight = 3.03` to address class imbalance (24.8% churn rate) and a tuned decision threshold of 0.40, found by maximizing F1 along the precision-recall curve on the test set. Two baselines are computed for comparison: a class-balanced **logistic regression** with standardized features, and a transparent **rule-based heuristic** (`NPS ≤ 4 OR complaints ≥ 2 OR days_since_topup > 10`). Training is fully supervised on the synthetic churn label.

Generative AI (transformer-based LLM, prompt-based — no fine-tuning). Three of the four required capabilities — natural-language segmentation, multi-channel content generation, and impact narration — are open-ended generation problems for which an LLM is the only practical fit. The prototype uses **Anthropic Claude Sonnet 4**, a transformer-based foundation model, through the official SDK. Each of the seven Claude routes runs a separate prompt chain with a strict output JSON schema; no fine-tuning or RAG store is used. Prompts include the relevant subscriber-aggregate statistics so generation is grounded in the data, not hallucinated.

Why not deep learning end-to-end? A neural sequence model on raw CDR could in principle learn richer representations, but the 18 hand-engineered features already capture the dominant signal; the data volume (10K rows, 95M at production) is well below the threshold where deep models start to win on tabular data; and interpretability matters for regulatory and CVM-team trust.

3.2 Technology stack

Layer	Technology	Rationale
Frontend	Next.js 14 (App Router), TypeScript, Tailwind CSS, Recharts, Framer Motion, Radix UI	Server components for fast initial render; typed component contracts; standard, production-ready stack
AI reasoning	Anthropic Claude Sonnet 4 via <code>@anthropic-ai/sdk</code>	State-of-the-art reasoning quality, native streaming, lowest latency among frontier models for this size class
ML model	XGBoost 2.x, scikit-learn (LR baseline, scaler, metrics)	Best-in-class tabular performance; mature scikit-learn ecosystem for evaluation
Data	Local CSV (10K subscribers); Postgres / Supabase planned for production	CSV is appropriate at prototype scale; the API layer abstracts the source so a swap is local
Integrations	Twilio Voice (<code>twilio</code> v6) for outbound retention calls; Nodemailer v8 with Gmail SMTP for retention emails	Already wired into the codebase via <code>/api/send-call</code> and <code>/api/send-email</code> ; activated by <code>TWILIO_SID</code> / <code>TWILIO_TOKEN</code> / <code>TWILIO_PHONE_FROM</code> and <code>GMAIL_ADDRESS</code> / <code>GMAIL_APP_PASSWORD</code>
Hosting	Vercel (serverless edge + Node runtime)	Zero-ops deploy from GitHub; preview deployments per branch
Versioning & CI	Git, GitHub	Standard

The full stack is summarized inside the Model Evaluation tab of the running app.

```

graph TD
    Automation[Automation]
    subgraph Automation_Box [Automation]
        A1[Host & VM Image Builder]
        A2[Terraform -> Network CIs]
        A3[Reduction -> Release Workflow -> Docker CI]
    end
    DevOps((DevOps)) --- Automation_Box
    CI_CD[CI/CD]
    subgraph CI_CD_Box [CI/CD]
        C1[api/clone]
        C2[api/prepare]
        C3[api/compile]
        C4[api/prepare]
        C5[api/package]
        C6[api/validate build]
        C7[api/build artifact]
        C8[api/build test]
        C9[api/feedback]
    end
    CI_CD_Box --> Artifact[Artifact]
    CI_CD_Box --> Deployed[Deployed]
    CI_CD_Box --> CI_CD_Box2[CI/CD]
    subgraph Artifact_Box [Artifact]
        A1a[Veracut (clap -> build)]
        A1b[GitHub Actions CI]
    end
    subgraph Deployed_Box [Deployed]
        D1[Twilio voice]
        D2[Heroku deploy SFTP]
    end
    subgraph CI_CD_Box2 [CI/CD]
        C10[Antigen, Cloud Foundry 4 (platform-as-a-service)]
        C11[Kubernetes 3.x (cloud-native platform)]
    end
    CI_CD_Box2 --> Data1[Data]
    CI_CD_Box2 --> Data2[Data]
    subgraph Data1_Box [Data]
        D1a[IBM Watson Analytics/IBM Watson WatsonCloud, stored data]
        D1b[Stored event artifact and data stored at s3.amazonaws.com]
    end
    subgraph Data2_Box [Data]
        D2a[Stored event artifact and data stored at s3.amazonaws.com]
    end

```

4. Prototype Development

4.1 Scope of this report

The prototype is a production-grade Next.js 14 web application deployed at telcopulse-ai.vercel.app. It exposes **nine dashboard modules** sharing one synthetic subscriber dataset and one trained model — Churn Radar, Subscriber Workflow, Smart Segments, Campaign Writer, Impact Predictor, What-If Simulator, Model Evaluation, Persona Architect, and Launch Copilot. A short description of each module, screenshots, and the live demo link live in the repository's [README.md](#).

Per the rubric's *narrowly scoped* requirement, **the rest of this report focuses on a single module — the **Subscriber Workflow**** — because that is where the prototype's value moment lives and where every AI Factory component (ML inference, generative AI, human-in-the-loop, feedback loop) is exercised in one screen. The eight other modules are supporting surfaces; treat them as breadth.

4.2 The Subscriber Workflow module — deep dive

Subscriber Workflow is a per-subscriber retention tracker with four explicit states (AI Scored → Reviewed → Contacted → Outcome) and a tightly designed interaction loop:

1. **AI Scored.** XGBoost has already scored every subscriber (batch). The expanded row shows the customer profile, the rule-based risk drivers, and the predicted probability with risk tier.
2. **Generate AI brief.** A single button calls `/api/subscriber-brief`. Claude Sonnet 4 returns a strict-schema JSON: a 2–3 sentence risk narrative citing the decisive features, a recommended action (channel + offer + urgency + rationale), and a fully drafted personalized email + Twilio voice script. A "Claude Sonnet 4" badge confirms the live call; a "Template fallback" badge appears if no API key is set.
3. **Reviewed — three differentiated decision paths.** The Marketer Decision panel is *gated behind the brief* so the analyst must read the model's reasoning before committing. The three buttons drive materially different paths:
 - **Approve AI** — accept the prediction. Contact step proceeds with Claude's recommended channel and offer.
 - **Escalate** — agree the subscriber is at risk and want a richer intervention. The contact step is shown with a red "Escalated to senior CVM agent" banner; the channel is auto-pinned to voice; the Email button is disabled.
 - **Mark Safe** — override the model as a false positive. The contact step is hidden, step 3 (CONTACTED) renders as (skipped), the outcome is auto-set to "No outreach needed", and the reviewer notes prompt switches to a guided placeholder asking *why* this was a false positive — the highest-value label for retraining.

- 4. **Contacted (skipped on Mark Safe).** Editable email subject + body and voice script. Send Email goes through `/api/send-email` (Gmail SMTP via Nodemailer); Voice Call goes through `/api/send-call` (Twilio TwiML). The send buttons use the *edited* values, not the original draft.
- 5. **Outcome.** Retained / Churned / No response, plus a free-text reviewer-notes textarea.
- 6. **Save feedback.** A single submit button POSTs the full row — predicted probability, risk tier, human action, channel used, outcome, notes, AI-brief flag, recommendation channel, ISO timestamp — through `/api/feedback`. The endpoint logs to the serverless function output (in production this becomes a Postgres / Supabase insert). The row is then locked, the four step pills turn green, and the workflow is observably closed.

Because this single module exercises ML inference, generative AI, three distinct HITL paths, real outbound channels, and the labelled-feedback sink, it is by design the report's only deep-dive surface.

4.3 The value moment

A CVM analyst opens **Churn Radar**, sees the top-N highest-risk subscribers ranked by predicted probability — the dashboard surfaces a concrete number on this screen ("Projected MRR loss if no action: **\$1,247**" against the six most-at-risk subscribers in the current synthetic dataset). The analyst opens one subscriber, clicks *Generate AI brief*, picks one of the three decision paths, edits the message if needed, dispatches it (or skips on Mark Safe), records the outcome, and clicks **Save feedback** — all from one tab, in under a minute. The same flow today involves three teams and two days.

4.4 Architecture in production

In production the same prototype would only require three changes: replace the local CSV with a managed Postgres / Supabase instance, replace the in-line model file with an inference endpoint (or re-export to ONNX for edge serving), and add per-tenant authentication. Every other layer — Next.js API routes, Claude prompt chains, the UI — remains as-is.

5. Success Metrics and Go / No-Go Evaluation

5.1 Prototype success metrics

Model performance (XGBoost, 2,000-row test set, F1-optimal threshold 0.40):

Metric	Value	Notes
ROC-AUC	0.73	Threshold-independent ranking quality
Recall (churn)	0.66	Two-thirds of true churners flagged
Precision (churn)	0.41	Low-cost FPs acceptable in CVM
F1-score	0.50	At F1-optimal threshold
Inference latency	0.3 ms / sample	Well under any real-time SLA
Confusion matrix	TN 1,023 · FP 480 · FN 168 · TP 329	

Baseline comparison (same test set):

Model	AUC	Precision	Recall	F1
XGBoost (primary)	0.73	0.41	0.66	0.50
Logistic Regression	0.72	0.39	0.66	0.49
Rule-based heuristic	0.69	0.49	0.55	0.52

XGBoost wins on AUC, the threshold-independent measure of how well the model ranks risky subscribers above safe ones. The rule-based baseline shows higher F1 at its single fixed cutoff, but its lower AUC means it cannot be tuned across operating points the way the model can.

Pipeline metrics. The training script runs end-to-end in under three seconds on a laptop, and the scoring step writes a 10K-row CSV in under one second. The code path has no manual steps and is one-command reproducible.

User interaction metric (target for pilot). Time from "open Churn Radar" to "send first retention campaign" — target ≤ 90 seconds for a CVM analyst, vs. the current state of ~2 days across three teams.

5.2 Outcome metrics

Outcome metric	Baseline	Target with model	Source
Subscribers correctly contacted / month	~700K (rule-based)	~2.2M (model recall × IOH base × churn rate)	Calculated
Subscribers retained / month	—	~549K (at 25% offer-acceptance benchmark)	Calculated
Revenue saved / month	—	~\$4.4M (at \$8.50 ARPU × partial 12-mo LTV)	Calculated
Annualized impact	—	~\$52.7M	Calculated
Analyst handle-time per retention case	~2 days	≤ 90 seconds	Workflow estimate

These figures are illustrative; production numbers depend on real production data quality and offer cost. The full breakdown is in the Business Impact panel of the prototype.

5.3 Go / No-Go criteria and decision

Targets are **recall-prioritized**: in retention, missing a true cherner costs lifetime ARPU; contacting a non-cherner costs a small offer. The asymmetry justifies a precision target of 0.40 rather than the more typical 0.60.

Metric	Target	Actual	Status
AUC-ROC	≥ 0.65	0.73	PASS
Recall	≥ 0.50	0.66	PASS
Precision	≥ 0.40	0.41	PASS
F1	≥ 0.45	0.50	PASS
Inference latency	< 5 s	0.3 ms	PASS

Risks identified through experimentation, baseline benchmarking, and operator domain knowledge.

- *Distribution drift*. Competitor mass-promo events compress usage signals and inflate false positives; projected AUC drop to ~0.60 in this regime.
- *Data quality at scale*. Missing NPS and corrupted tenure post-SIM-re-registration are known production issues.
- *Generative AI hallucination*. Claude-generated content must be brand-reviewed before first send; current prototype does not enforce a human approval gate on outbound copy.
- *Precision floor*. At 0.41 precision, 59% of flagged subscribers are false positives, which is acceptable for a low-cost SMS but would not be for a costly hardware-subsidy offer.

Recommendation: GO with conditions. All five thresholds pass on the synthetic test set. Three pre-conditions before scaled rollout:

1. **Data-quality sprint.** Unify legacy subscriber IDs across CRM and billing; backfill 30 days of missing NPS via median imputation plus an `nps_missing` indicator feature; correct tenure for re-registered SIMs.
2. **Real-data recalibration.** Retrain on the most recent 90 days of production CDR / billing / NPS. Expected AUC range: 0.65–0.75 (telecom industry norm).
3. **A/B pilot.** 10,000 subscribers split 50/50 between model-targeted and random outreach for two consecutive 30-day cycles. Promote to full production only if (a) churn reduction ≥ 10% with $p < 0.05$, and (b) offer-cost-to-revenue-saved ratio ≤ 0.30. If either condition fails, fallback to **Conditional Go**: deploy only on the highest-decile risk segment (where precision is materially higher) and revisit broader rollout after one quarter.

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